

Anthony Avila

05/01/2026

Observation Paper

Prior to undergrad, I worked as a professional musician for four years. The guitar and the digital audio workstation (DAW) were my instruments of choice, and as time went on I focused more on audio production. One of the core skills of audio production is sound design, in which one creates unique and novel sounds. Subtractive synthesis is the most commonplace strategy for many producers, including myself. It entails creating a synthesizer patch by starting with a basic waveform (sine, square, triangle, sawtooth, pulse-width) and filters out frequencies with pass-filtering and audio effects; typically "noisier" waveforms with more harmonics are preferred so that there are frequencies to subtract. It is this simple strategy that most popular electronic music genres design their sounds with.

One day, many years ago, I was creating a synth patch from scratch with headphones on (typically, I work alone in a silent room), and I ended up holding down a key that played the default raw sawtooth waveform on my synthesizer for too long. When I stopped holding down the key, I noticed an effect on my hearing as I listened to my breathing that could be described as sounding like running water. Perhaps more accurately, it sounded like a chorus effect (i.e., multiple duplicate voices of a frequency, all with slight de-tunings, which is a common effect used in indie rock for its dreamy character) or low-quality audio warping algorithms (e.g., 0.5x speed on YouTube) applied to a sound, but only for maybe one second at most before it went away.

To determine the effect boundary, I built a small browser app and ran four self-test rounds developed serially and ad-hoc, rating each preset as perceived, boundary, or not perceived. Every round used a 10-second white-noise "chaser" played immediately after the test tone, except for one silence-control trial in round 1. I found that the effect was not perceivable without background noise available to be "distorted" by the effect. Round 1 focused on a wide breadth of parameters tested using

saw, sine, and square waves at frequencies from 80 Hz to 3000 Hz across 1, 5, and 30-second durations. Sine never elicited an aftereffect. Square waves elicited the effect roughly as often as saw. Sawtooth was tested at different durations (1 second failed, 5 seconds worked, 30 seconds was strongest); and the silence-chase control failed entirely. Rounds 2 through 4 fixed tone duration at 10 seconds and varied harmonic density, which I define here as the fraction of total integer harmonics needed to approach a true saw or square wave. Round 2 tested full and half density of saw and square at 20, 200, and 400 Hz: every 20 Hz preset failed, while 200 was perceived regardless of density and 400 Hz roughly denoted a perceived boundary. Rounds 3 and 4 stepped down to three-eighths, quarter, and eighth density at 60, 200, and 450 Hz, and revealed the same asymmetry: square continued to elicit the effect at lower densities than saw and across a wider frequency range. Overall, this experiment allowed me to generally define some of the boundary conditions of the effect, but with some redundancy in variables due to the ad-hoc nature of the development of each round in the app.

A few concepts of psychoacoustics explain most of what these rounds revealed. The stimuli first: a sawtooth wave contains every integer harmonic of its fundamental, a square wave only the odd-numbered harmonics, and a sine wave none. The perceptual signature created by this harmonic weighting is what Wolfe et al. (2018, pp. 286–287) call timbre.

The aftereffect itself is best explained by neural adaptation. Neurons that respond to a sustained stimulus reduce their firing rate while it occurs and recover only gradually once it stops (Wolfe et al., 2018). Cochlear and downstream channels tuned to the tone's harmonics adapt during the tone, and for a brief window after, those channels are under-responsive relative to baseline neighbors. The aftereffect is the perceptual signature of that imbalance, and round 1's duration sweep backs it directly, since longer sawtooth tones produced stronger aftereffects, which is what the dose-response curve adaptation predicts.

The manner in which the cochlea encodes the frequencies that adapt is not one mechanism but actually three. At low fundamentals, individual auditory-nerve fibers fire in step with the rising phase of the wave, a behavior called phase locking that supports a precise temporal code up to roughly 1 kHz

(Wolfe et al., 2018, pp. 299–300). Past that band, the refractory period of a single fiber gets in the way, so the system shifts to the volley principle, in which small groups of fibers take turns firing on alternating cycles such that their collective output still tracks the wave out to about 4 or 5 kHz (Wolfe et al., 2018, p. 300). Above 5 kHz, temporal coding effectively disappears and pitch is carried only by which points along the basilar membrane are mechanically displaced, since each location is tuned to its own frequency (Wolfe et al., 2018, pp. 294–295, 301–302, 351). The rounds line up with this picture. 200 and 400 Hz sit deep inside the phase-locking band and were perceived everywhere; 3000 Hz, where the temporal code is breaking down, failed in round 1; 20 Hz, whose harmonics fall on the apex of the cochlea where place resolution and sensitivity are weakest, failed throughout round 2; and pure sine, lacking the extra harmonics needed to recruit additional channels, never elicited the effect.

The “chaser” audio matters for a separate reason. White noise spreads its energy evenly across the audible range (Wolfe et al., 2018, p. 306), and when noise follows a tone, the auditory system fills the post-tone interval with content extrapolated from the preceding tone, an illusion Wolfe et al. (2018, pp. 342–343) call perceptual restoration. The silence chaser in round 1 failed because there was nothing to restore into. The “running water” quality reads as that restored content, shaped by the channel imbalance that neural adaptation left behind.

The waveform asymmetry in rounds 3 and 4 reflects another rule: low-frequency content masks high-frequency content asymmetrically, what Wolfe et al. (2018, pp. 306–307) call upward spread of masking. Sawtooth's lower partials drown out its upper partials as density falls; a square wave keeps only the odd harmonics, so its partials are more widely separated along the basilar membrane and suffer less mutual masking.

My hypothesis is that harmonic density, not parity (i.e., all integer harmonics versus odd-integer harmonics), drives this aftereffect, and the square-versus-saw asymmetry at low density follows from upward spread of masking rather than from anything special about total versus odd harmonics. The prediction is that saw and square should be indistinguishable at full density and diverge as density falls,

with square decaying more slowly because its partials are spaced more widely along the basilar membrane. Rounds 2 through 4 lean this way, with full density giving saw and square equivalent results and lower densities favoring square. The perceived effect strength is also possibly positively correlated with duration, as was strongly observed in round 1, but is beyond the scope of the remainder of the paper.

Assuming infinite resources, I would test this hypothesis with a single experiment, using a two-dimensional grid varying which harmonics (the individual frequency components that make up a complex sound) are retained. Axis 1 is a membership rule: every-Nth integer harmonic for $N = 1$ (sawtooth) through 8; "odd only" (square); "even only"; "primes only"; "powers of two"; and 4 additional stochastic random subsets matched in count. Axis 2 is density: $1/2^N$ of the harmonics for $N = 1, 2, 4, 8, 16, 32$, and the full set. These axes represent my independent variable (the harmonic rule and effective density of the tone), while my dependent variable is the reported strength of the perceived aftereffect. Each cell is synthesized at three fundamentals (70 Hz, 200 Hz, 500 Hz) that were inferred from self-reported perceived boundaries from the app, 10 seconds of tone followed by 10 seconds of white-noise chaser, with levels equalized along ISO 226:2003 equal-loudness contours (Wolfe et al., 2018, pp. 303–304). The listening population would consist of 4 groups: trained audio engineers, non-musicians, and their respective controls. The independent variable is hypothesized to change the dependent variable. That is, as harmonic density decreases, the aftereffect strength decreases uniformly, with at most a small horizontal shift attributable to upward spread of masking differences, where lower-frequency content drowns out higher-frequency content.

Several alternative explanations need ruling out before claiming a perceptual mechanism. Headphone resonance at the tested fundamentals could create the effect as a transducer artifact rather than an auditory-system effect. Tinnitus-style aftereffects unrelated to the stimulus and attentional priming from rating the same self-selected illusion across repeated trials are plausible counterexplanations.

Bibliography

Wolfe, J. M., Kluender, K. R., Levi, D. M., Bartoshuk, L. M., Herz, R. S., Klatzky, R. L., & Merfeld, D. M. (2018). *Sensation & perception* (5th ed.). Oxford University Press.

Avila, A. (2026). Running water audio illusion ad hoc diagnostic app [Computer software]. GitHub. <https://github.com/anthony-a-avila/running-water-audio-illusion-ad-hoc-diagnostic-app>